McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Lab Project: Project Proposal
Crossing Gate System

Instructor: Rameez Ashraf

SMRTTECH 3CC3

Cloud Computing and Internet of Things

B. Tech - SMRTTECH

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Lab Section: L04

Abstract

There are many intersections between railroads and cars. To ensure the safety of the passengers in the cars, automated gate systems must be implemented to tell cars to stop before the intersection when a train is coming. There should also be a sensor to detect if a car or person or any other obstruction is mistakenly in the intersection even when the gates are down. This project aims to achieve this task by building a prototype of this system using a microcontroller, ultrasonic sensor, infrared sensors, and servo motors. This project intends to close the gates to an intersection when sensors detect a train is coming, turn on an obstruction detection system once it knows a train is coming, turn off the obstruction detection system once the train arrives, then open the gates to the intersection when the train has left. If the obstruction detection system detects an obstruction, the system intends to send this information to the cloud so that it will alert the train to slow down or stop if necessary.

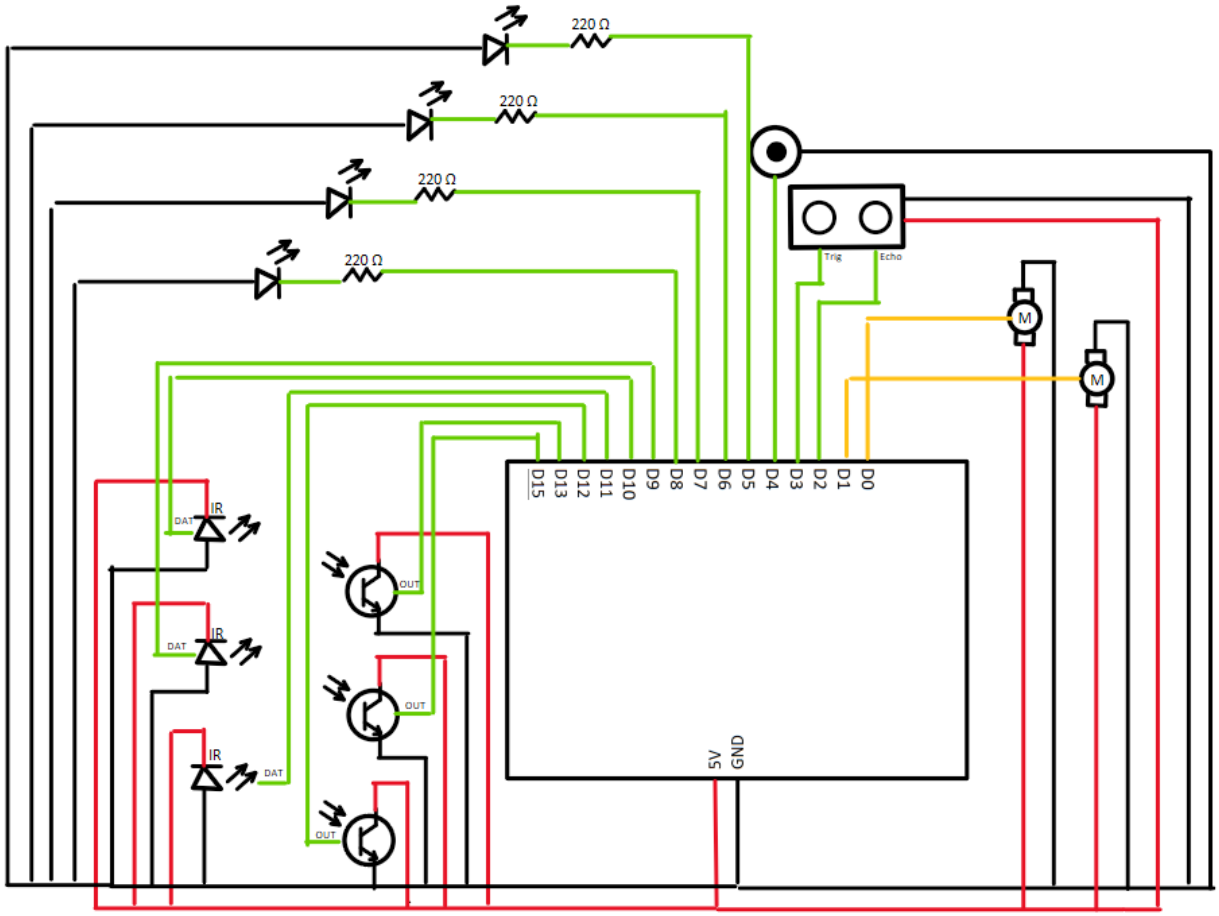
Description

In countries all over the world, there are intersections where railroads for freight cargo trains or public transportation trains pass through roads for civilian automobiles. A safety system must be implemented to ensure the safety of the civilians whenever a train passes, and because there could be many of these intersections for one railroad, automating this safety system is paramount for efficient safety. Our automated crossing gate system employs sensors and actuators to achieve the task of closing and opening gates on both sides of the railroad to block civilian cars from crossing the railroad when a train is passing by. This is done by 3 pairs of IR transmitters and receivers and 2 servo motors. The first IR sensor pair detects a train is coming from kilometres away, activating the 2 servo motors to close 2 gates on both sides of the railroad. The second IR sensor pair is past the railroad and detects when the train has fully passed the crossing, so that the gates can be reopened and the civilian cars can drive through. The third IR sensor pair is in conjunction with 1 ultrasonic sensor as a subsystem to detect if there are current obstructions on the crossing. The ultrasonic sensor has a threshold of a minimum distance, and if it detects less than that distance, it knows something is blocking it, and that there is a car or another obstruction in the crossing. If so, it will send this data to the cloud, and if the train is connected to the cloud, it will receive an alert so that it can slow down or stop completely. The third IR sensor pair's purpose is to detect if the train is nearby enough to the intersection and if so, it will disable the ultrasonic sensor system so that it does not mistake the train as an obstruction.

List of Components

- 1 Arduino MEGA
- 1 ultrasonic sensor
- 3 pairs of IR transmitters + receivers
- 2 servo motors
- 4 red LEDs
- 4 resistors
- 1 passive buzzer
- Breadboards and wires
- PC with LabVIEW and Ubidots
- Props (toy car, toy train)
- Cardboard

Electrical Schematic



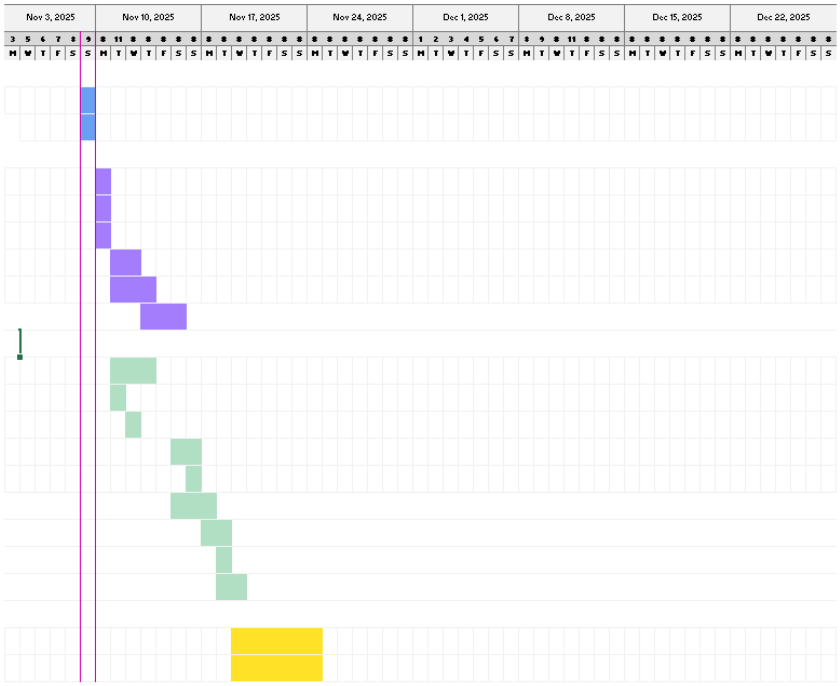
Gantt Chart Timeline

RAILROAD CROSSING GATE Project start: 9-Nov-25

Jacob Kwan 400453713

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Display week 0



TASK	ASSIGNED TO	PROGRESS	START	END
Initiation				
Gather cardboard	Nicolas	0%	11-9-25	11-9-25
Buy IR sensors	Nicolas	0%	11-9-25	11-9-25
Planning and design				
Fill in task breakdown & GANTT chart	Jacob	0%	11-10-25	11-10-25
Write abstract & project description	Nicolas	0%	11-10-25	11-10-25
Draw electrical schematic	Jacob	0%	11-10-25	11-10-25
Figure out IR sensor connections	Jacob	0%	11-11-25	11-12-25
Write psuedocode for high level logic of system	Jacob	0%	11-11-25	11-13-25
Write code in C++ in Arduino IDE	Jacob	0%	11-13-25	11-15-25
Execution				
Build railroad crossing environment and gates	Nicolas	0%	11-11-25	11-13-25
Paint/draw details on cardboard	Nicolas	0%	11-11-25	11-11-25
Set up sensors and motors in position	Nicolas	0%	11-12-25	11-12-25
Connect Arduino IDE to LabView	Jacob & Nicolas	0%	11-15-25	11-16-25
Build HMI for system	Jacob & Nicolas	0%	11-16-25	11-16-25
Convert logic in C++ to LabView block diagrams	Jacob & Nicolas	0%	11-15-25	11-17-25
Connect LabView to Ubidots	Jacob & Nicolas	0%	11-17-25	11-18-25
Ensure connection is working	Jacob & Nicolas	0%	11-18-25	11-18-25
Set up Ubidots dashboard and data tracking	Jacob & Nicolas	0%	11-18-25	11-19-25
Troubleshooting				
Final inspection of system	Jacob & Nicolas	0%	11-19-25	11-24-25
Troubleshooting and testing	Jacob & Nicolas	0%	11-19-25	11-24-25

STAGE	MILESTONES	TASKS	OWNER	TIME ALLOTTED
1	All materials on hand	- Gather cardboard - Buy IR sensors	Nicolas	3 days
2	Project proposal completed	- Fill in task breakdown & GANTT chart - Write abstract and description - Draw electrical schematics	Nicolas & Jacob	1 day
3	Model railroad crossing constructed	- Build railroad crossing environment and gates - Paint/draw details on cardboard - Set up sensors and motors in appropriate areas	Nicolas	1 day
4	Simulated code generated in Arduino IDE	- Figure out how IR sensors work - Write psuedocode for high level logic of the entire system - Write the code in C++ in the Arduino IDE	Jacob	2 days
5	LabView code completed	- Figure out how to connect Arduino IDE to LabView - Build HMI for the system - Convert logic in C++ to LabView block diagrams	Nicolas & Jacob	2 days
6	Ubidots connection completed	- Connect LabView to Ubidots - Ensure connection is working - Set up Ubidots dashboard and data tracking	Nicolas & Jacob	1 day
7	Code is thouroughly tested	- Final inspection of system - Troubleshooting and testing	Nicolas & Jacob	3 days